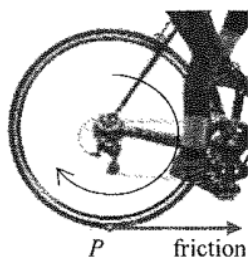


1. (a)	Heat given out by iron cube = Heat gained by water $0.2 \times 450 \times (T - 33) = 0.6 \times 4200 \times (33 - 25)$ $T = 257 (^{\circ}\text{C})$	2M	
		1A	
		3	
(b)	Some water might become steam (immediately). <u>Or</u> Some water in the cup might splash.	1A	
		1A	
		1	
(c)	It is because some heat is lost to the surroundings through vaporization of water / as latent heat. <u>Or</u> It is because some heat is lost to the surroundings (or polystyrene cup) / The water might not be well stirred, otherwise, the water should have reached a temperature higher than 33°C (i.e. underestimated).	1A	
		1A	
		1A	
		1A	
		2	
(d) (i)	Poor thermal contact between the glass bob of the thermometer and the iron cube.	1A	
		1	
(ii)	Infra-red (IR) thermometer.	1A	
		1	

2. (a) $m_0 = \text{density} \times \text{volume}$ $= 1.20 \times (1.5 \times 10^{-3})$ $= 1.8 \times 10^{-3} \text{ kg or } 1.8 \text{ g}$	1M 1A 2	
(b) (i) $p_1 V = n_1 R T$ and $p_2 V = n_2 R T$ As $m_0 \rightarrow 4m_0$, $n_2 = 4n_1$ $\therefore p_2 = \frac{n_2 R T}{V} = \frac{4n_1 R T}{V} = 4p_1 = 4.0 \times 10^5 \text{ Pa (or } 4.0 \text{ atm)}$ <u>Or</u> $pV = nRT$ $p \propto n$ (as V , R and T are constant) Since $m_0 \rightarrow 4m_0$, thus $n \rightarrow 4n$, p becomes 4 times the original value (= 1.0 atm), $p = 4.0 \times 10^5 \text{ Pa (or } 4.0 \text{ atm)}$	1M 1A 1M 1A 2	
(ii) More air molecules hit unit area of surface of wall or bottle / water surface (as $n \rightarrow 4n$ moles of molecules trapped in the same volume), The surface of wall or bottle / water surface is hit more frequently OR rate of change of total momentum, i.e. force, per unit area of wall or bottle / water surface increases, pressure increases.	1A 1A 2	
(iii) $\Delta F = \text{increase in pressure} \times \text{area}$ $= [(4 - 1) \times 10^5] \times 0.014$ $= 4200 \text{ N}$	1M 1A 2	error-carried-forward (e.c.f.) from (b)(i)
(c) (i) The pressure force acting on the water results in its change of momentum (a water jet). Magnitude of the force = rate of change of momentum. According to Newton's third law of motion, an equal but opposite force / reaction / thrust acts on the rocket. (According to Newton's second law of motion,) the reaction force causes the rocket to accelerate / move upwards.	1A 1A 2	
(ii) kinetic energy of (the jet of) water / work done on (the jet of) water	1A 1	

3. (a)



1A

1

- (b) Mechanical power delivered (for work done against the resistive force)

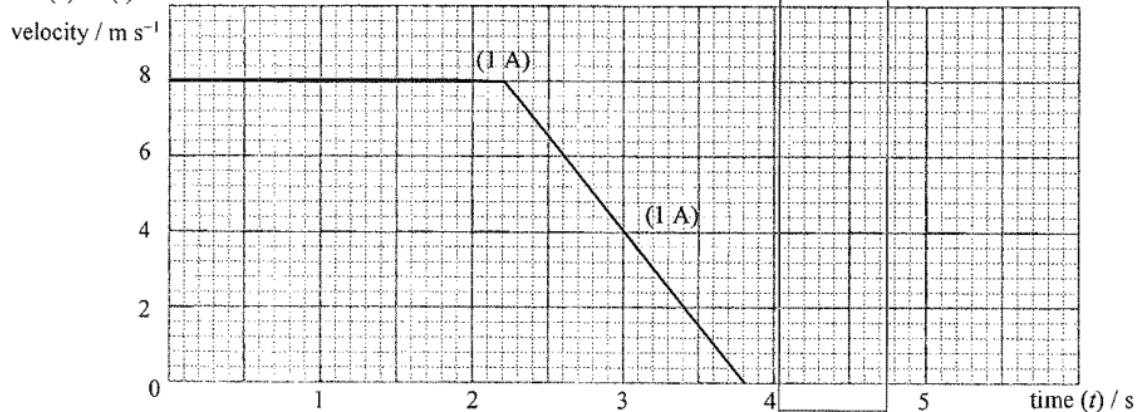
$$\begin{aligned}
 P &= Fv \\
 &= 17.0 \times 8.0 \\
 &= 136 \text{ W}
 \end{aligned}$$

1M

1A

2

- (c) (i)



2

- (ii) Deceleration: $\left| \frac{0-8}{3.8-2.2} \right| = 5.0 \text{ m s}^{-2}$
 Resistive force = 65×5.0
 $= 325 \text{ N}$

1M

e.c.f. from (c)(i)

1A

Alternative method:

$$\text{Braking distance} = \frac{1}{2}(8)(3.8 - 2.2) = 6.4 \text{ m}$$

$$Fs = \frac{1}{2}mv^2 - 0$$

$$F(6.4) = \frac{1}{2}(65)(8)^2$$

$$F = 325 \text{ N}$$

1M

1A

2

- (iii) Total distance travelled after $t = 2.0 \text{ s}$
 $=$ area under the graph from $t = 2.0 \text{ s}$ to $t = 3.8 \text{ s}$
 $= \frac{0.2+1.8}{2} \times 8.0 = 8.0 \text{ m}$
 Distance between the bicycle and the obstacle when the bicycle stops = $9 - 8$
 $= 1 \text{ m}$

1M/1A

1M

1A

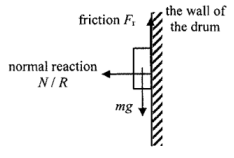
3

- (d) The padded cushion can increase the contact time of the knee and the ground such that a smaller force acts on the knee.

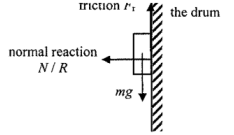
1A

1A

2

4. (a) (i)		2A	
			2
(ii) Normal reaction (force acting on drum/wall of the drum by the object)		1A	
			1
(b) (i)	$\omega = 78.5 \text{ rad s}^{-1}$ rotation speed = $\frac{78.5}{2\pi} \times 60$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$ Or $T = \frac{2\pi}{\omega} = 0.0800441 \text{ s}$ and $f = \frac{1}{T} = 12.493663 \text{ rev s}^{-1}$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$	1M	
		1A	
		1M	
		1A	
			2
(ii) centripetal force	$= m\omega^2 r$ $= (2 \times 10^{-5})(78.5)^2(0.25)$ $= 0.03081125 \text{ N} \approx 0.0308 \text{ N}$	1M	
		1A	
			2
(iii) $v = r\omega$	$= 0.25 \times 78.5$ $= 19.625 \approx 19.6 \text{ m s}^{-1}$	1M	
		1A	
			2
(iv) 1		1A	
			1

61

		2A	
			2
(ii) Normal reaction (force acting on drum/wall of the drum by the object)		1A	
			1
(b) (i)	$\omega = 78.5 \text{ rad s}^{-1}$ rotation speed = $\frac{78.5}{2\pi} \times 60$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$ Or $T = \frac{2\pi}{\omega} = 0.0800441 \text{ s}$ and $f = \frac{1}{T} = 12.493663 \text{ rev s}^{-1}$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$	1M	
		1A	
		1M	
		1A	
			2
(ii) centripetal force	$= m\omega^2 r$ $= (2 \times 10^{-5})(78.5)^2(0.25)$ $= 0.03081125 \text{ N} \approx 0.0308 \text{ N}$	1M	
		1A	
			2
(iii) $v = r\omega$	$= 0.25 \times 78.5$ $= 19.625 \approx 19.6 \text{ m s}^{-1}$	1M	
		1A	
			2
(iv) 1		1A	
			1

5. (a) (i) $a / i / q$

1A

1

(ii) $c / k / s$

1A

1

(b) (i) wavelength (4 grids) = 4×50 cm
= 200 cm or 2 m

1A

1

(ii) speed = frequency $\times$ wavelength
frequency = speed $\div$ wavelength
= $340 \div 2$
= 170 Hz

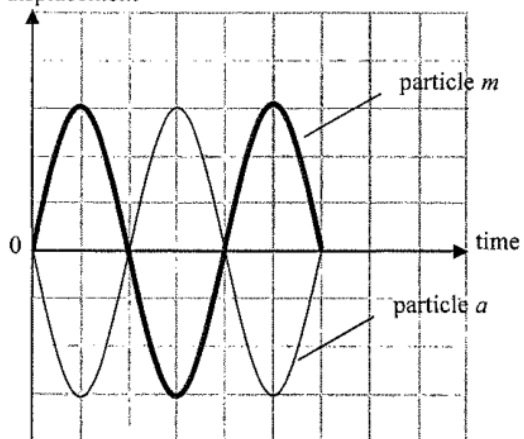
1M

1A

2

(iii)

displacement



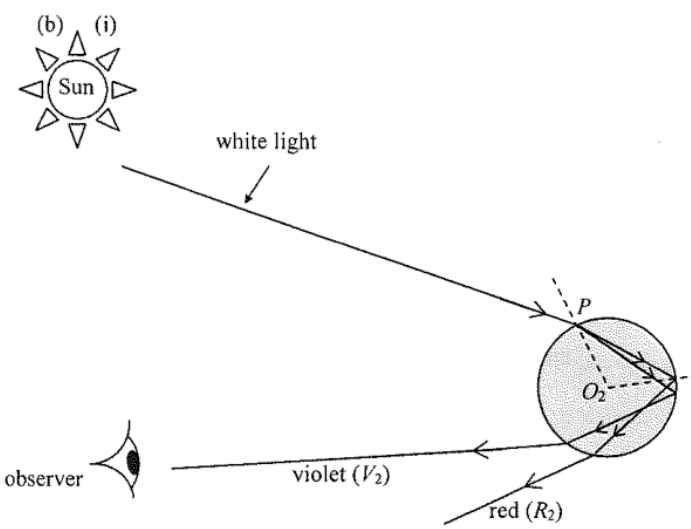
2A

2

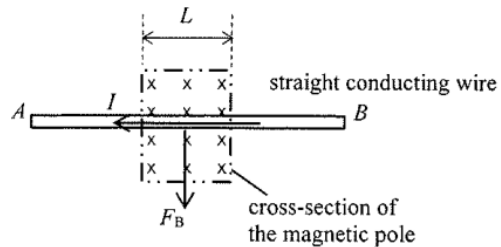
(c) Displacement increases, i.e. further to the left.

1A

1

Solution	Marks	Remarks
6. (a) (i) $\sin r = \frac{\sin 45^\circ}{1.325}$ $r = 32.253454^\circ \approx 32.3^\circ$	1M 1A	
(ii) $c = \sin^{-1} \frac{1}{1.325}$ $c = 49.000646^\circ \approx 49.0^\circ$ $r \text{ (or } \angle OBA) = 32.3^\circ < c = 49.0^\circ$, therefore total internal reflection does not occur at point B.	2 1M 1A 1A 3	
(b) (i)  The diagram shows a sun emitting a beam of white light towards a spherical water droplet. The light enters the droplet at point P. Inside the droplet, the light is dispersed into its constituent colors, with violet light (labeled V₂) being refracted more than red light (labeled R₂). An observer is shown on the left, looking towards the droplet. The center of the droplet is labeled O₂. The light rays are shown as solid lines, and the normal at point P is shown as a dashed line.	2A 2	
(ii) There is more light energy absorbed by water droplet due to a longer path. <u>Or</u> There is more light energy transmitted to air at the point of reflection as the light undergoes reflection twice.	1A 1A 1A 1A 2	

7.	(a)	$R = \rho \frac{l}{A} \Rightarrow 0.50 = \rho \frac{0.20}{8.0 \times 10^{-7}}$ $\rho = 2.0 \times 10^{-6} (\Omega \text{ m})$	1M	
			1A	
				2
	(b)	(i) Terminal 3	1A	
				1
		(ii) When switch S is connected to terminal 2, the equivalent resistance across MN		
			1M	
			1A	
				2
		(iii) (p.d. across R_2 is) greater than (p.d. across R_1)	1A	
				1
		(iv) zero as no p.d. across CF / no current in the coil.	1A	
			1A	
				2

Solution		Marks	Remarks													
8.	(a)		1A													
			1													
	(b)		3A													
			3													
	(c)	<table><tr><td></td><td>magnetic force F_B</td><td></td></tr><tr><td>using a stronger magnet of the same dimensions</td><td>increase</td><td>1A</td></tr><tr><td>using another support such that the position of the horizontal wire is slightly lowered</td><td>unchanged</td><td>1A</td></tr><tr><td>using a longer wire and the current is kept unchanged</td><td>unchanged</td><td>1A</td></tr></table>		magnetic force F_B		using a stronger magnet of the same dimensions	increase	1A	using another support such that the position of the horizontal wire is slightly lowered	unchanged	1A	using a longer wire and the current is kept unchanged	unchanged	1A		
		magnetic force F_B														
	using a stronger magnet of the same dimensions	increase	1A													
	using another support such that the position of the horizontal wire is slightly lowered	unchanged	1A													
	using a longer wire and the current is kept unchanged	unchanged	1A													
			3													
(d)	(i)	(Switch on the power source.) Adjust the power source to deliver a small current, say, 0.5 A to the wire and record the ammeter and the balance readings. Adjust the power source to increase the current stepwise (0.5 A) and record the corresponding balance reading.	1A 1A													
			2													
	(ii)	The balance reading increases linearly with the current.	1A													
			1													
	(iii)	The graph does not pass through the origin because the balance has not been reset to zero initially before the experiment.	1A	Accept: because of the weight of the support / wire												
			1													

9.	(a)	(i)	alpha particle / α	1A	
				1	
	(ii)	6 α -decays, 4 β -decays	$4 n_{\alpha} = 232 - 208$ $2 n_{\alpha} + (-1) n_{\beta} = 90 - 82$	1A	
				1A	
	(b)	$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}} = \left(\frac{1}{2}\right)^{\frac{10}{1.41 \times 10^{10}}}$ $\text{Portion} = 1 - \frac{N}{N_0} = 1 - \left(\frac{1}{2}\right)^{\frac{10}{1.41 \times 10^{10}}}$ $= 4.9159 \times 10^{-10} \approx 4.9 \times 10^{-10} \text{ (2 sig. fig.)}$	$N = N_0 e^{-\lambda t} \quad \text{and} \quad \lambda = \frac{\ln 2}{t_{1/2}}$ $\lambda = \frac{\ln 2}{t_{1/2}} = 4.9159 \times 10^{-11} \text{ yr}^{-1}$ $1 - \frac{N}{N_0} = 1 - e^{-\frac{\ln 2}{t_{1/2}} t}$ $= 1 - e^{-\frac{\ln 2}{1.41 \times 10^{10}} (10)}$ $\approx 4.9 \times 10^{-10}$	2	
				1M	
				1A	
				2	
	(c)	(i)	Aluminium body blocks alpha and/or beta particles emitted from thoriated glass.	1A	
				1A	
		(ii)	Thorium-232 has an extremely long half-life.	1	
				1A	
			<u>Or</u> As part (b) suggests, only an extremely small fraction (4.9×10^{-10}) of thorium decaying in 10 years.	1A	
				1	